

Learning a preference relation with MacSum Choquet Integrals

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Abstract

Multicriteria decision support systems often rely on the use of a particular set function, a capacity, which values the influence for the decision of not just a single criterion, but also a coalition of criteria. From a dataset of examples that is used to find a suitable capacity, it is possible to mimic the decision-making behaviour of one or more experts. However, it is often more reliable to ask an expert to give a preference between two examples rather than an overall score for all training examples, especially when the examples are given sequentially. In existing works, approaches have been proposed to tackle this problem but they are often based on constraints on the capacity (k -additivity, sparsity,...). In this paper, we introduce the use of a recent generalization of possibility measures called Macsum to model preferences. A Macsum model can be seen as a convex set of linear models whose number of parameters equals the number of criteria - like linear models - and which intrinsically respects game modelling. This model can be trained from a dataset by means of a classical gradient descent. We propose a novel approach to train a Macsum online from preferences given sequentially.

1 Introduction

We place ourselves in a context where examples (or cases, or individuals) are described by means of values associated to criteria (or variables, attributes, features). Multicriteria decision support systems are efficient tools to give suitable recommendations among cases to users based on the aggregation of values for the given set of criteria [Roy, 1996]. Such an aggregation can be done by means of weights, or user's preferences, or importance, or influence of each criterion or of a coalition of criteria.

The weighted sum aggregation are known to be insufficient to represent all preference relations. Although, theses preferences is often represented as a capacity [Grabisch, 2016], a set function that associates each subset of criteria to a value. Then, more sophisticated aggregation functions based on capacity like Choquet integrals [Grabisch and Labreuche, 2010] to represent the synergies between criteria are more suitable.

One challenge in multicriteria decision support is to elicit a correct capacity to reflect the user's preferences and many authors have taken an interest in learning capacity [Lesca and Perny, 2010; Benabbou *et al.*, 2017]. One way to obtain such a capacity is to learn it from alternatives given as a dataset of pairwise comparisons among examples. An alternatives between a pair of examples may highlight a preference of one example over the other, or it may also be an indifference among them. Hence two possibilities may be considered. In the first one, an history of examples an the associated decision is given [Fürnkranz and Hüllermeier, 2010]. In the second one, examples of preferences are provided sequentially or online [Zhao *et al.*, 2016].

The definition of a capacity relies on the determination of a high number of values as it should provide each subset of a set of criteria to a value (among 2^n values to find for a set of n criteria). Thus, some hypotheses need to be made because of this numerous number of parameters to learn.

In [Herin *et al.*, 2024], an approach based on the learning of the Möbius transform of a capacity has been proposed. It is based on the use of a Choquet integral defined from the Möbius transform of the unknown capacity. A first algorithm proposed in [Herin *et al.*, 2024] doesn't take into account any constraints of monotonicity on the Möbius transform and some approximations are made because the number of parameters is important. This algorithm is based on a the Regularized Dual Averaging (RDA) online algorithm [Xiao, 2009]. However, it requires the setting of several hyper-parameters and necessitates a high number of training examples to offer decent results.

Another parametric model, the so-called MacSum, has been proposed in [Strauss *et al.*, 2022]. This model is based on a pair of Choquet integrals and is able to represent input/output models. It is an imprecise linear model whose simplicity of representation enables it to be used in a same way that a linear model. The learning of the MacSum parameters, considering a data set of pairs of input/output, mimics the regression analysis and used gradient descent [Hmidy *et al.*, 2023].

Moreover, a MacSum model provides an imprecise linear representation: it captures, in the imprecision outputted by the model, the difficulty of representing preferences by a linear model. Like the linear model, it can assign negative weights to certain criteria, making it possible to take into ac-

count both monotonicity of preference with regard to certain criteria and anti-monotonicity with regard to others. This aspect is an important issue as a linear model could represent this by assigning a negative weight to such a criterion. On the contrary, a model based on capacities doesn't allow us to represent the fact that a criterion could have a negative impact on preferences.

As the MacSum-based model is less sophisticated than a capacity-based model, it is not as good at representing complex situations involving synergy between criteria. However, it has many advantages over the capacity-based model. The number of parameters required is equal to n the number of criteria, whereas the capacity-based model requires 2^n parameters. Being monotonic with respect to some inputs and anti-monotonic with respect to others, monotonicity property doesn't have to be constrained during training.

In this paper, a new approach to learn a MacSum algorithm from online pairs of preferences among example is introduced. This algorithm is compared to the Choquet integral learning online algorithm proposed by [Herin *et al.*, 2024].

This article is organised as follows. Section 2 recalls the background: notations, set functions, and Choquet integrals. A brief reminder of the learning algorithm by [Herin *et al.*, 2024] and of the MacSum model used in this article is also presented. Section 3 presents theoretical results concerning the parameters of MacSum which enable us to learn them. In this section, a presentation is done on how to model the given preference relation. Section 4 introduces our proposed adaptation of the MacSum learning algorithm to be able to be trained from online pairs of preferences. In Section 5, an experiment with the proposed algorithm is presented and a comparison with the [Herin *et al.*, 2024]'s algorithm is done. Section 6 concludes the paper and presents some future work.

2 Background

This section presents the technical results used in this article. We start with the notations, next we present the set functions and particular cases: the capacities, the Möbius inverse, the Choquet integrals, online algorithms to learn the capacities and MacSum aggregation.

2.1 Notations

We place ourselves in the multicriteria decision making (MCDM) context where objects or alternatives are evaluated according to n criteria, or attributes, from a set $\mathcal{N} = \{1, \dots, n\}$. These criteria are used to represent object from a given set $\mathcal{X}$. Every object $x \in \mathcal{X}$ is represented by a vector $\mathbf{x} = (x_1, \dots, x_n)$ where x_i represents the evaluation of x with respect to the criterion i . All criterion values are expressed in the same scale L . This scale can be $L = [0, 1]$, as in [Herin *et al.*, 2024], or it can also be more general and set to $L = \mathbb{R}$. This last case is more convenient when considering non monotonic Choquet integral. Throughout this paper, $L = \mathbb{R}$ and it is denoted equivalently either L , or $\mathbb{R}$. In this setting, a vector $\mathbf{x} \in \mathbb{R}^n$ is associated to 2 vectors of $\mathbb{R}^n$ $\mathbf{x}^+$ and $\mathbf{x}^-$ such that for all $i \in \mathcal{N}$, $x_i^+ = \max(0, x_i)$ and $x_i^- = \min(0, x_i)$.

2.2 Set Functions and Möbius Inverse

In MCDM, importance of subsets of criteria from $\mathcal{N}$ and their interactions can be modelled with a capacity $\mu : 2^{\mathcal{N}} \rightarrow [0, 1]$. A capacity is an increasing set function μ [Grabisch, 2016], that is $\mu(A) \leq \mu(B)$ for all $B \subseteq A \subseteq \mathcal{N}$, and such that $\mu(\emptyset) = 0$ and $\mu(\mathcal{N}) = 1$. When the capacity μ is also additive (i.e. when $\mu(A \cup B) = \mu(A) + \mu(B)$ for any $A, B \subseteq \mathcal{N}$ such that $A \cap B = \emptyset$) then it corresponds to a probability measure on $\mathcal{N}$.

A model based on capacities doesn't allow us to represent the fact that a criterion could have a negative impact on preferences. This is the case, for example, with drug interactions: drugs taken separately can be highly effective in treating pathologies, but their combination can make them dangerous to health.

The Choquet integral (see Section 2.3) is an aggregation function defined by means of a set function. It can be expressed with the Möbius inverse of a set function [Grabisch, 2016]. Namely, the Möbius inverse (or Möbius for short in this paper) of a set function μ is the set function $m^\mu : 2^{\mathcal{N}} \rightarrow \mathbb{R}$ defined from μ as:

$$m^\mu(A) = \sum_{B \subseteq A} (-1)^{|A \setminus B|} \mu(B), \text{ for any } A \subseteq \mathcal{N}. \quad (1)$$

Note that for each $i \in \mathcal{N}$, we have $m^\mu(\{i\}) = \mu(\{i\})$.

From the Möbius m^μ it is possible to deduce the set function μ : for any $A \subseteq \mathcal{N}$, $\mu(A) = \sum_{B \subseteq A} m^\mu(B)$.

The learning of a Möbius can be an alternative to learn a capacity or a set function [Herin *et al.*, 2024]. In this case, it is often associated with a k -additivity property for m^μ (ie. $m^\mu(A) = 0$ for any $A \in \mathcal{N}$ such that $|A| \geq k$ for a given integer $k > 0$) in order to reduce the number of values to learn.

In the following, in order to be friendly reading, a capacity is denoted v and μ is dedicated to a more general set functions.

2.3 Choquet Integral

The Choquet integral [Grabisch, 2016] is defined for any vectors $\mathbf{x}$ with values in $\mathbb{R}$ (or any interval of $\mathbb{R}$ such as $[0, 1]$) and enables to aggregate the values of $\mathbf{x}$ according to a capacity v that takes into account interactions between criteria. The Choquet integral is widely used in MCDM, see for instance [Chateaufneuf, 1994] or [Grabisch and Labreuche, 2010].

The classic definition of the Choquet integral uses an ascending ranking of the values of the vector $\mathbf{x} = (x_1, \dots, x_n)$:

$$C_v(\mathbf{x}) = \sum_{i=1}^n (x_{(i)} - x_{(i-1)}) v(A_{(i)}) \quad (2)$$

where $()$ is the permutation on $\mathcal{N}$ such that $x_{(1)} \leq \dots \leq x_{(n)}$ and $A_{(i)} = \{(i), \dots, (n)\} \subseteq \mathcal{N}$, with the convention $x_{(0)} = 0$.

Let us present some important remarks concerning Choquet integral:

- When the capacity is additive the Choquet integral is an expectation.

• Choquet integral can be defined with respect to any set function (and not necessarily a capacity) with the only condition of being zero for the empty set. Such a set function is also called a game [Grabisch, 2016] and may not be increasing or even monotonic (see [Murofushi *et al.*, 1994; De Waegenaere and Wakker, 2001] for more details).

• The Choquet integral can be expressed with the Möbius of a set function μ (here μ can be either a capacity or a more general set function that simply complies with $\mu(\emptyset) = 0$) [Chateauneuf and Jaffray, 1989; Grabisch, 2016]:

$$C_\mu(\mathbf{x}) = \sum_{A \subseteq \mathcal{N}} m^\mu(A) \cdot \min_{i \in A} x_i. \quad (3)$$

Equation (3) highlights the fact that the Choquet integral is the inner product $\langle \mathbf{m}, \phi(\mathbf{x}) \rangle$ of two vectors of $2^{\mathcal{N}}$: $\mathbf{m}$ and $\phi(\mathbf{x})$, the components of these two vectors being defined for an arbitrary order of the 2^n subsets of $\mathcal{N}$: $2^{\mathcal{N}} = \{A_1, \dots, A_{2^n}\}$ by $\forall k = 1, \dots, 2^n, m_k = m^\mu(A_k)$ and $\phi_k(\mathbf{x}) = \min_{i \in A_k} x_i$.

Hence the Choquet integral seems similar to the multilinear model used in multi-attribute utility theory [Dyer and Sarin, 1979] and in game theory [Owen, 1972]. This remark enables to construct the online algorithm presented in the next section.

2.4 Online Learning of a Set Function

This part briefly recalls the online algorithm proposed in [Herin *et al.*, 2024] to learn the Möbius vector $\mathbf{m}$ associated to a capacity v when the Choquet integral C_v is used for modelling a preference relation.

Online preference learning works sequentially [Hoi *et al.*, 2021]. At each round t , a pair of criteria vectors $(\mathbf{x}_t, \mathbf{x}'_t)$ is available to update the vector $\mathbf{m}$ such that either $\mathbf{x}_t \succ \mathbf{x}'_t$ or $\mathbf{x}_t \sim \mathbf{x}'_t$, otherwise $\mathbf{x}_t$ and $\mathbf{x}'_t$ are exchanged.

The batch counterpart problem may be considered, i.e., we assume to have a training set of preference pair of examples $(\mathbf{x}_t, \mathbf{x}'_t)_{t \in \mathcal{T}}$ where the set of all rounds, denoted by $\mathcal{T}$, is divided into two parts:

- $\mathcal{P}$ the set of indices corresponding to the strict preference: $\mathbf{x}_t \succ \mathbf{x}'_t$,
- $\mathcal{I}$ the set of indices corresponding to the indifference: $\mathbf{x}_t \sim \mathbf{x}'_t$.

At each round t , the current model (the learner) m_t makes a prediction. In case of error the learner suffers a loss $l_t(m_t)$, with $l_t : \mathcal{M} \rightarrow \mathbb{R}$ a loss function defined on the set of models $\mathcal{M}$. The aim of the online learning algorithm is to minimize the cumulative error on the preference examples combined with a regulation term [Tsochantaridis *et al.*, 2005; Waegeman *et al.*, 2009; Herin *et al.*, 2023].

The method is based on the following parameters to express how the indifference is considered:

- $\delta > 0$ a threshold used to separate preference to indifference;
- positive error variables: ϵ_t , ϵ_{t+} and ϵ_{t-} , $t \in T$, making flexible strict preference and indifference.

The constraints associated are:

$$\begin{aligned} C_v(\mathbf{x}_t) - C_v(\mathbf{x}'_t) &\geq \delta - \epsilon_t && \text{if } \mathbf{x}_t \succ \mathbf{x}'_t \text{ i.e., } t \in \mathcal{P} \\ C_v(\mathbf{x}_t) - C_v(\mathbf{x}'_t) &\leq \delta + \epsilon_{t+} && \text{if } \mathbf{x}_t \sim \mathbf{x}'_t \text{ i.e., } t \in \mathcal{I}, \\ C_v(\mathbf{x}'_t) - C_v(\mathbf{x}_t) &\leq \delta + \epsilon_{t-} && \text{if } \mathbf{x}_t \sim \mathbf{x}'_t \text{ i.e., } t \in \mathcal{I}. \end{aligned}$$

The regularized loss function, to minimize is

$$\frac{1}{T} \sum_{t=1}^T (l_t(m_t) + \lambda \Psi(m_t))$$

where:

- parameter λ is used to control the level of regularisation,
 - the loss function at the round t is defined as
$$l_t(m_t) = \begin{cases} [\delta - C_v(\mathbf{x}_t) + C_v(\mathbf{x}'_t)]^+ & \text{if } t \in \mathcal{P} \\ [C_v(\mathbf{x}_t) - C_v(\mathbf{x}'_t) - \delta]^+ & \text{if } t \in \mathcal{I}. \end{cases}$$
 - the sparse inducing penalty used in [Herin *et al.*, 2024] is defined with the L^1 -norm: $\Psi(m) = \|\mathbf{m}\|_1$ to favour sparse solutions (ie. solutions with a high number of null components).
- A remark on this approach is: in case of error from the learning, this function is not equal to 0:
- if $t \in \mathcal{P}$ the loss function is not equals to 0 if $\delta - C_v(\mathbf{x}_t) + C_v(\mathbf{x}'_t) > 0$ i.e., $\mathbf{x}_t \sim \mathbf{x}'_t$ considering the constraints associated.
 - if $t \in \mathcal{I}$ the lost function is not equals to 0 if $C_v(\mathbf{x}_t) - C_v(\mathbf{x}'_t) - \delta > 0$ i.e., $\mathbf{x}_t \succ \mathbf{x}'_t$ considering the constraints associated.

2.5 RDA Algorithm for Capacity Learning

In [Herin *et al.*, 2024], a Regularized Dual Averaging (RDA) algorithm [Xiao, 2009] is introduced to learn a Möbius transform instead of the classical Online Gradient Descent (OGD). In a RDA algorithm, at the round $t + 1$, the model m_{t+1} that minimize the cumulative regularized loss for all the past rounds is considered unlike in the OGD algorithm where only the past round t is considered [Shalev-Shwartz, 2012].

In [Herin *et al.*, 2024], it is proven that the equation obtained for the minimisation needs to be solve with a quadratic program at each criterion but we can bound the loss function by another one (since l_t is convex) and the obtained equation admit an efficient closed-form solution:

$$m_{t+1} = -\frac{\sqrt{t}}{\gamma} (|\bar{\mathbf{g}}_t| - \lambda)_+ \text{sign}(\bar{\mathbf{g}}_t).$$

where

- for any sequence of vectors $\mathbf{x}_1, \dots, \mathbf{x}_t, \bar{\mathbf{x}}_t = \frac{1}{t} \sum_{\tau=1}^t \mathbf{x}_\tau$,
- for any vector $\mathbf{x}$, the vector $\text{sign}(\mathbf{x})$ is defined as: for $i \in \mathcal{N}$, $\text{sign}(\mathbf{x})_i = 1$ if $x_i > 0$, $\text{sign}(\mathbf{x})_i = -1$ if $x_i < 0$ and $\text{sign}(\mathbf{x})_i = 0$ if $x_i = 0$,
- considering $t \in \mathcal{T}$, $\mathbf{g}_t$ is the vector defined as follows: $\phi(\mathbf{x}_t) = (\phi_S(\mathbf{x}_t), S \subseteq \mathcal{N})$ with $\phi_S(\mathbf{x}_t) = \min_{i \in S} (x_t)_i$, where v_t is the set function at the round t :
 - if $t \in \mathcal{P}$ then $\mathbf{g}_t = -(\phi(\mathbf{x}_t) - \phi(\mathbf{x}'_t))$ if $C_{v_t}(\mathbf{x}_t) - C_{v_t}(\mathbf{x}'_t) < \delta$ and $\mathbf{g}_t = 0$ otherwise.

- if $t \in \mathcal{I}$ then $\mathbf{g}_t = -(\phi(\mathbf{x}_t) - \phi(\mathbf{x}'_t))$ if $C_{v_t}(\mathbf{x}_t) - C_{v_t}(\mathbf{x}'_t) < -\delta$, $\mathbf{g}_t = \phi(\mathbf{x}_t) - \phi(\mathbf{x}'_t)$ if $C_{v_t}(\mathbf{x}_t) - C_{v_t}(\mathbf{x}'_t) > \delta$, and $\mathbf{g}_t = 0$ otherwise.

2.6 MacSum Aggregation

In this paper, our aim is to propose to learn a set function using the MacSum aggregation which is defined using a pair of parametric Choquet integrals. Our motivation is to learn an imprecise model but with a number of parameters to be determined that equals the number of criteria $|\mathcal{N}|$ instead of the number of criteria subsets $2^{\mathcal{N}}$ (or less when a k -additivity has to be set to a low value for k).

The MacSum operator is a set function parametrised by a vector $\varphi \in \mathbb{R}^n$ considered as a function $\varphi : \mathcal{N} \rightarrow \mathbb{R}$. In the following, to lighten the notations, for all $i \in \mathcal{N}$, $\varphi(i)$ is denoted by φ_i . In the same way as for $\mathbf{x}$, we define the vector φ^+ (resp. φ^-) by for all $i \in \mathcal{N}$, $\varphi_i^+ = \max(\varphi_i, 0)$ (resp. $\varphi_i^- = \min(\varphi_i, 0)$). These 2 vectors φ^+ and φ^- are used to defined the MacSum set function μ_φ and its conjugate μ_φ^c :

- the MacSum set function $\mu_\varphi : 2^{\mathcal{N}} \rightarrow \mathbb{R}$ is $\mu_\varphi(A) = \max_{i \in A} \varphi_i^+ + \min_{i \in \mathcal{N}} \varphi_i^- - \min_{i \in A^c} \varphi_i^-$ for all $A \subseteq \mathcal{N}$, $A \neq \emptyset$, and $\mu_\varphi(\emptyset) = 0$.
- The conjugate of μ_φ is: $\mu_\varphi^c(A) = \mu_\varphi(\mathcal{N}) - \mu_\varphi(A^c)$ $\forall A \subseteq \mathcal{N}$ i.e., $\mu_\varphi^c(A) = \min_{i \in A} \varphi_i^- + \max_{i \in \mathcal{N}} \varphi_i^+ - \max_{i \in A^c} \varphi_i^+$.

Note that we have¹ $\mu_\varphi^c \leq \mu_\varphi$ that entails for all $\mathbf{x} \in \mathcal{X}$, $C_{\mu_\varphi^c}(\mathbf{x}) \leq C_{\mu_\varphi}(\mathbf{x})$.

The MacSum set function μ_φ and its conjugate μ_φ^c are used to define MacSum aggregation $\mathcal{A}_{\mu_\varphi}(\mathbf{x})$ of any $\mathbf{x} \in \mathcal{X}$ as

$$\mathcal{A}_{\mu_\varphi}(\mathbf{x}) = [C_{\mu_\varphi^c}(\mathbf{x}), C_{\mu_\varphi}(\mathbf{x})].$$

It should be noted that the MacSum aggregation represents a convex set of linear models.

Considering permutations associated to vector φ , MacSum aggregation have the following expressions [Hmidy *et al.*, 2023]

$$C_{\mu_\varphi}(\mathbf{x}) = \sum_{k=1}^n \varphi_{[k]}^+ \cdot \left(\max_{i=1}^k x_{[i]} - \max_{i=1}^{k-1} x_{[i]} \right) + \sum_{k=1}^n \varphi_{[k]}^- \cdot \left(\min_{i=1}^k x_{[i]} - \min_{i=1}^{k-1} x_{[i]} \right), \quad (4)$$

$$C_{\mu_\varphi^c}(\mathbf{x}) = \sum_{k=1}^n \varphi_{[k]}^+ \cdot \left(\min_{i=1}^k x_{[i]} - \min_{i=1}^{k-1} x_{[i]} \right) + \sum_{k=1}^n \varphi_{[k]}^- \cdot \left(\max_{i=1}^k x_{[i]} - \max_{i=1}^{k-1} x_{[i]} \right), \quad (5)$$

where

- $[\cdot]$ is a permutation that sorts φ in decreasing order $\varphi_{[1]} \geq \varphi_{[2]} \geq \dots \geq \varphi_{[n]}$, with $\varphi_{[n+1]} = \varphi_{[n]}$,

¹ A partial order $\leq$ between capacities defined on the universe $\mathcal{N}$ is usually defined as: $\mu \leq \mu'$ iff $\mu(A) \leq \mu'(A)$ for all $A \subseteq \mathcal{N}$.

- $[\cdot]$ is a permutation that sorts φ in increasing order $\varphi_{[1]} \leq \varphi_{[2]} \leq \dots \leq \varphi_{[n]}$ with $\varphi_{[n+1]} = \varphi_{[n]}$,
- $\max_{i=1}^0 x_{[i]} = 0 = \min_{i=1}^0 x_{[i]}$,
- $\min_{i=1}^0 x_{[i]} = 0 = \max_{i=1}^0 x_{[i]}$.

Learning MacSum aggregation is similar to learning linear function with regression (see [Hmidy *et al.*, 2023] for more details).

3 MacSum Aggregation: Theoretical Results

In this section, the theoretical foundations of the proposed MacSum online algorithm are introduced.

3.1 Learning the Parameters of MacSum

First let us note that the parameters of MacSum aggregation could be calculated using Möbius transform. Let m^{μ_φ} (resp. $m^{\mu_\varphi^c}$) be the Möbius associated to the MacSum set function μ_φ (resp. μ_φ^c).

Property 1. For all $i \in \mathcal{N}$, $m^{\mu_\varphi}(i) > 0$ if and only if $m^{\mu_\varphi}(i) = \varphi_i^+ > 0$.

Proof. We have $\forall i$, $\mu_\varphi(i) = m^{\mu_\varphi}(i)$ and $\mu_\varphi(i) = \varphi_i^+ + \min_{i \in \mathcal{N}} \varphi_i^- - \min_{j \neq i} \varphi_j^-$, hence $m^{\mu_\varphi}(i) = \varphi_i^+ + \min_{j \neq i} \varphi_j^- - \min_{i \in A^c} \varphi_i^-$ which entails

- $m^{\mu_\varphi}(i) > 0$ corresponds to $\varphi_i > 0$,
- $m^{\mu_\varphi}(i) = 0$ corresponds to $\varphi_i \leq 0$ and φ_i^- is not the unique minimum value of φ .
- $m^{\mu_\varphi}(i) < 0$ correspond to φ_i is the unique minimum value of φ .

So the φ_i such that $\varphi_i > 0$ are equal to the coefficient $\mu^\varphi(i)$ that are strictly positive. $\square$

We obtain similar results for the negative values of MacSum set function.

Property 2. For all $i \in \mathcal{N}$, $m^{\mu_\varphi^c}(i) < 0$ if and only if $m^{\mu_\varphi^c}(i) = \varphi_i^- < 0$.

Proof. $\forall i \in \mathcal{N}$, $m^{\mu_\varphi^c}(i) = \varphi_i^- + \max_{i \in \mathcal{N}} \varphi_i^+ - \max_{j \neq i} \varphi_j^+$ so we have

- $m^{\mu_\varphi^c}(i) < 0$ corresponds to $\varphi_i < 0$,
- $m^{\mu_\varphi^c}(i) = 0$ corresponds to $\varphi_i \geq 0$ and φ_i^+ is not the unique maximum value of φ .
- $m^{\mu_\varphi^c}(i) > 0$ correspond to φ_i is the unique maximum value of φ .

So, the algorithm 1 proposed by [Herin *et al.*, 2024] enables to calculate the positive parameters of MacSum aggregations. Hence negative parameters can be calculated in two ways:

1) calculating the Möbius transform of μ_φ^c using the relation proved in [Grabisch, 2016]:

$$m^{\mu_\varphi^c}(A) = (-1)^{|A|+1} \sum_{B|A \subseteq B} m^{\mu_\varphi}(B) \forall A \subseteq \mathcal{N}.$$

2) using the following property: for all $\mathbf{x} \in [0, 1]^n$ we have $C_\mu(\mathbf{x}) = 1 - C_{\mu^c}(\mathbf{1} - \mathbf{x})$. So we may compute $\mu_{\mathcal{T}}^c$ considering the preference $(\mathbf{1} - \mathbf{y}, \mathbf{1} - \mathbf{x})$. with the [Herin *et al.*, 2024]’s algorithm 1.

However, in this case, we are making approximation assumptions that are useless to us. In fact it is possible to prove that the bounds of MacSum aggregation have an inner product representation that enables us to learn the parameters without approximation hypothesis.

Property 3. *The Choquet integrals $C_{\mu_\varphi}(\cdot)$ and $C_{\mu_\varphi^c}(\cdot)$ have an inner product representation: there exists $\bar{\phi}(\mathbf{x})$ and $\underline{\phi}(\mathbf{x})$ such that $C_{\mu_\varphi}(\mathbf{x}) = \langle \varphi, \bar{\phi}(\mathbf{x}) \rangle$ and $C_{\mu_\varphi^c}(\mathbf{x}) = \langle \varphi, \underline{\phi}(\mathbf{x}) \rangle$.*

Proof.

Let us define $\bar{\mathbf{y}}$ and $\underline{\mathbf{y}}$ as follows.

Let, $\forall k \in [1, n]$,

$$\bar{y}_k^+ = (\max_{i=1}^k x_{[i]} - \max_{i=1}^{k-1} x_{[i]}),$$

$$\bar{y}_k^- = (\min_{i=1}^k x_{[i]} - \min_{i=1}^{k-1} x_{[i]}),$$

$$\underline{y}_k^+ = (\min_{i=1}^k x_{[i]} - \min_{i=1}^{k-1} x_{[i]}) \text{ and}$$

$$\underline{y}_k^- = (\max_{i=1}^k x_{[i]} - \max_{i=1}^{k-1} x_{[i]}).$$

Let σ denote the bijection such that $\forall k \in [1, n]$, $\sigma([k]) = k$ and remark that $[k] = [n - k + 1]$.

Now let $\bar{y}_k = \bar{y}_{\sigma(k)}^+$ if $\varphi_{[k]} \geq 0$ and $\bar{y}_{\sigma(n-k+1)}^-$ else. Let

$\underline{y}_k = \underline{y}_{\sigma(k)}^+$ if $\varphi_{[k]} \geq 0$ and $\underline{y}_{\sigma(n-k+1)}^-$ else. Now let $\bar{\phi}$ be

the application that associates $\bar{\mathbf{y}}$ to $\mathbf{x}$ and $\underline{\phi}$ be the application

that associates $\underline{\mathbf{y}}$ to $\mathbf{x}$, by construction, we have: $C_{\mu_\varphi}(\mathbf{x}) =$

$$\langle \varphi, \bar{\phi}(\mathbf{x}) \rangle \text{ and } C_{\mu_\varphi^c}(\mathbf{x}) = \langle \varphi, \underline{\phi}(\mathbf{x}) \rangle.$$

□

3.2 Preference Relation Associated to MacSum

In this paper, considering the MacSum preference relation associated to the parameter φ , knowing whether $\mathbf{x}$ is preferred to $\mathbf{x}'$ or not, is obtained by comparing the middle of the intervals $\mathcal{A}_{\mu_\varphi}(\mathbf{x})$ and $\mathcal{A}_{\mu_\varphi}(\mathbf{x}')$. This value is denoted

$$\mathcal{M}(\mathbf{x}, \varphi) = \frac{1}{2} \cdot (C_{\mu_\varphi^c}(\mathbf{x}) + C_{\mu_\varphi}(\mathbf{x})).$$

With property 3, we have: $\mathcal{M}(\mathbf{x}, \varphi) = \langle \varphi, \frac{1}{2} \cdot (\bar{\phi}(\mathbf{x}) + \underline{\phi}(\mathbf{x})) \rangle$. To simplify the notation, we denote: $\phi(\mathbf{x}) = \frac{1}{2} \cdot (\bar{\phi}(\mathbf{x}) + \underline{\phi}(\mathbf{x}))$.

Thus, since $\mathcal{M}(\mathbf{x}, \varphi)$ can be expressed by a scalar product, its derivative with respect to φ is simply the vector $\phi(\mathbf{x})$.

4 Learning MacSum for a Preference Relation

The aim of this section is the presentation of our algorithm to learn a preference relation using MacSum.

The principle of the learning method we propose consists in minimizing a quadratic norm, each time t a pair of values $\mathbf{x}_t, \mathbf{x}'_t$ is presented, modifying the value of the φ weight vector if the prediction is incorrect.

Starting from a random vector φ_0 , the method consists in using a gradient descent approach:

$$\varphi_{t+1} = \varphi_t - \beta \cdot \Gamma(\mathbf{x}_t, \mathbf{x}'_t),$$

where β is a learning factor and $\Gamma(\mathbf{x}_t, \mathbf{x}'_t)$ is the derivative of the loss function with respect to φ .

Algorithm 1 MacSum Quadratic learning

Input: pairwise examples $(\mathbf{x}_t, \mathbf{x}'_t)_{t \in \mathcal{T}}$, $\mathbf{x}_t \in [0, 1]^n$

Parameter: β

Output: φ

```

1:  $t \leftarrow 1, \varphi_k \leftarrow$  random signed value, with  $k \in \{1, \dots, n\}$ 
2: while no convergence do
3:   while  $t \leq T$  do
4:     receive pairwise example  $(\mathbf{x}_t, \mathbf{x}'_t)$ 
4:     compute loss gradient  $\Gamma(\mathbf{x}_t, \mathbf{x}'_t)$ 
4:      $\varphi_{t+1} \leftarrow \varphi_t - \beta \cdot \Gamma(\mathbf{x}_t, \mathbf{x}'_t)$ 
4:      $t \leftarrow t + 1$ 
5:   end while
6: end while
7: return  $\varphi_T$ 
```

For each round t , we consider that either $\mathbf{x}_t \succ \mathbf{x}'_t$ (i.e. $\mathbf{x}_t$ is preferred to $\mathbf{x}'_t$), we then say that $t \in \mathcal{P}$ (preference), or that $\mathbf{x}_t \sim \mathbf{x}'_t$ (i.e. no preference between $\mathbf{x}_t$ and $\mathbf{x}'_t$), we then say that $t \in \mathcal{I}$ (indifference).

To an input $\mathbf{x}_t$ and a weight φ_t we associate a score $\mathcal{M}(\mathbf{x}_t, \varphi_t)$.

We define a discrimination parameter $\delta \geq 0$ whose role is to distinguish between preference and indifference.

If $\mathbf{x}_t \succ \mathbf{x}'_t$ then we should have $\mathcal{M}(\mathbf{x}_t, \varphi_t) - \mathcal{M}(\mathbf{x}'_t, \varphi_t) \geq \delta$. If $\mathbf{x}_t \sim \mathbf{x}'_t$ then we should have $|\mathcal{M}(\mathbf{x}_t, \varphi_t) - \mathcal{M}(\mathbf{x}'_t, \varphi_t)| \leq \delta$.

To take account of this proposal, we propose the following quadratic criterion as the loss function:

$$\begin{aligned} \mathcal{L}(\mathbf{x}_t, \mathbf{x}'_t) &= \left([\delta - \mathcal{M}(\mathbf{x}_t, \varphi_t) + \mathcal{M}(\mathbf{x}'_t, \varphi_t)]^+ \right)^2 \text{ if } t \in \mathcal{P}, \\ &= \left([|\mathcal{M}(\mathbf{x}_t, \varphi_t) - \mathcal{M}(\mathbf{x}'_t, \varphi_t)| - \delta]^+ \right)^2 \text{ if } t \in \mathcal{I}, \end{aligned}$$

where $\forall u \in \mathbb{R}$, $[u]^+ = \max(u, 0)$.

- Interpretation of this loss function:

If $t \in \mathcal{P}$, i.e. $\mathbf{x}_t \succ \mathbf{x}'_t$ then the loss function is 0 if $\mathcal{M}(\mathbf{x}_t, \varphi_t) - \mathcal{M}(\mathbf{x}'_t, \varphi_t) < \delta$ else the loss function equals $(\delta - \mathcal{M}(\mathbf{x}_t, \varphi_t) + \mathcal{M}(\mathbf{x}'_t, \varphi_t))^2$.

If $t \in \mathcal{I}$, i.e. $\mathbf{x}_t \sim \mathbf{x}'_t$ then the loss function is 0 if $|\mathcal{M}(\mathbf{x}_t, \varphi_t) - \mathcal{M}(\mathbf{x}'_t, \varphi_t)| < \delta$ else the loss function equals $(|\mathcal{M}(\mathbf{x}_t, \varphi_t) - \mathcal{M}(\mathbf{x}'_t, \varphi_t)| - \delta)^2$.

- Deriving $\mathcal{L}_t(\mathbf{x}_t, \mathbf{x}'_t)$ with respect to φ leads to the following.

$$\text{Let } \Delta(\mathbf{x}_t, \mathbf{x}'_t) = \mathcal{M}(\mathbf{x}_t, \varphi_t) - \mathcal{M}(\mathbf{x}'_t, \varphi_t).$$

- if $t \in \mathcal{P}$ then,

$$\Gamma(\mathbf{x}_t, \mathbf{x}'_t) = 2 \cdot (\phi(\mathbf{x}_t) - \phi(\mathbf{x}'_t)) \cdot (\Delta(\mathbf{x}_t, \mathbf{x}'_t) - \delta)$$

if $\Delta(\mathbf{x}_t, \mathbf{x}'_t) \leq \delta$ and 0 else.

- if $t \in \mathcal{I}$ then,

$$\Gamma(\mathbf{x}_t, \mathbf{x}'_t) = 2 \cdot \text{sign}(\Delta(\mathbf{x}_t, \mathbf{x}'_t)) \cdot (\phi(\mathbf{x}_t) - \phi(\mathbf{x}'_t))$$

if $|\Delta(\mathbf{x}_t, \mathbf{x}'_t)| \geq \delta$, and 0 else,

where $\forall u \in \mathbb{R}$ $\text{sign}(u) = 1$ if $u \geq 0$ and -1 else.

Finally, this leads us to introduce Algorithm 1 to learn MacSum parameters.

Table 1: Accuracy comparison for different values of T .

T	700	200	3500	5000	7000
Herin et al.	0.0	0.0	0.716	0.717	0.726
MacSum	0.826	0.824	0.826	0.825	0.829

5 Experiments

In this section, we compare the approach proposed in [Herin et al., 2023] with our own by means of a first set of experiments.

To this end, we generate the 2^n coefficients of a capacity v on the one hand, and T pairs of alternatives $x_t, x'_t \in [0, 1]^n$ on the other. We then compute, for each pair (x_t, x'_t) , the values $y_t = C_v(x_t)$ and $y'_t = C_v(x'_t)$. We compare each pair and reorder them so that if $|y_t - y'_t| > \delta$ x_t is preferred to x'_t , that is, $y_t - y'_t > \delta$ (and thus $t \in \mathcal{P}$). If $|y_t - y'_t| \leq \delta$, then $x_t \sim x'_t$ (and thus $t \in \mathcal{I}$).

Since our approach does not take into account the fact that the Moebius transformation produces a monotonic set function (i.e., a capacity), we compare our approach with the unconstrained approach of [Herin et al., 2023] (Algorithm 1 in [Herin et al., 2023]). For the latter, we used the values given in the article, i.e. $\delta = 0.0001$, $\lambda = 0.1$ and $\gamma = 1000$. For our algorithm, we used the following parameter $\beta = 10^{-5}$. To detect the convergence of our algorithm, we calculate the average of the loss function over a test set (different from the learning set) and compare it with a threshold ϵ , which in this case is set to $\epsilon = 10^{-4}$.

In Table 1, an accuracy comparison is presented when the value of T varies. In this experiment, n is set to 5, and the accuracy is valued on a reference test set of 3000 examples, not used during the training. From this set, 3000 pairs are randomly chosen to evaluate the accuracy of the approaches.

There are two important differences between the method of [Herin et al., 2023] and the one we propose. The first of these differences is that, in the [Herin et al., 2023] algorithm, pairs are presented only once, whereas in our approach they are presented until the algorithm converges. This difference means that the [Herin et al., 2023] method converges faster when T is small. On the other hand, that leads to an accuracy highly dependent on the number of examples, as can be seen in Table 1. On the contrary, the MacSum algorithm shows more robust accuracy results for any size of training set. The second difference is that the complexity of each step is in n in the case of our method, whereas it is in 2^n in the case of the [Herin et al., 2023]’s algorithm.

6 Conclusion

In this article, we propose an alternative approach to that of [Herin et al., 2024] for learning a preference relation from a set of sequential examples. This approach is based on a recent generalization of possibility measure called Macsum. Macsum is a set function that can be seen as a convex set of linear relations, whose number of parameters equals the number of criteria - like a linear model - and which intrinsically respects game modelling. Such a model makes it possible to use a classic quadratic regression method. Since Macsum’s

number of parameters is equal to the number of criteria, its learning does not suffer from the curse of dimension. The first experiments we have conducted show that this method is able to model a preference relationship that would have been obtained using a monotonic model (Choquet integral with respect to a capacity). Its performance in terms of accuracy is even superior to a capacity-based approach.

In future work, more experiments will be conducted to better highlight the effect of its parameters in the final accuracy it provides. Moreover, the influence of pairs of indifferent examples will be studied deeper, for instance by studying the variation of the accuracy when the ratio of indifferent examples in the training set increases. Another promising work will be to focus on the use of Macsum output intervals in the learning algorithm, for instance by introducing interval comparison methods from the imprecise probability theory. Finally, it would be interesting to see how this method performs when preference relationships are not monotonic with respect to criteria.

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